

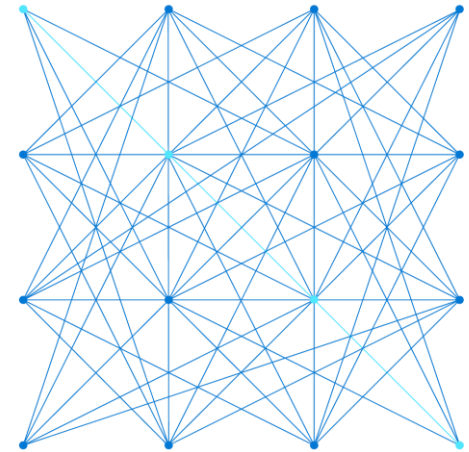
INSTRUCTOR-LED COURSE

DESIGN AND MANAGE ANALYTICS SOLUTIONS USING POWER BI

Mohammad Sbeeh

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www.sbeeh.com



We'll begin soon 



About me

MOHAMMAD SBEEH

Managing & Technical Director
Microsoft Certified Trainer (MCT)



Founder SBEEH Software
www.sbeeh.com

Certifications

MCP, MCTS, MCPD, MCITP, MCT & C# Corner MVP



20+ Years Experience

Solutions Architect & technical lead in software development.

Services: Digital Transformation | Software Consulting | Training | Outsourcing

Follow me on [Linkedin.com/in/sbeeh](https://www.linkedin.com/in/sbeeh)

Student Introductions

Name / Role

Company / Affiliation

Experience with data analytics and
Power BI

Expectations for the course



HANDS-ON COURSE

Master Power BI from Zero to Dashboard

Build interactive reports, model data with DAX, and turn raw numbers into decisions — in just 5 days.



What you'll learn

01

Connect & Transform

Import from Excel, SQL & web. Clean data with Power Query.

02

Data Modeling

Build relationships, star schemas & a solid data model.

03

DAX Formulas

Measures, calculated columns & time-intelligence.

04

Reports & Sharing

Interactive visuals, dashboards & publishing to the service.

Logistics

Class Schedule

Classroom requirements

Classroom etiquette / expectations

Building details (hours, parking, restrooms)



10 Minutes Assessment

Pre-assessment



Introduction

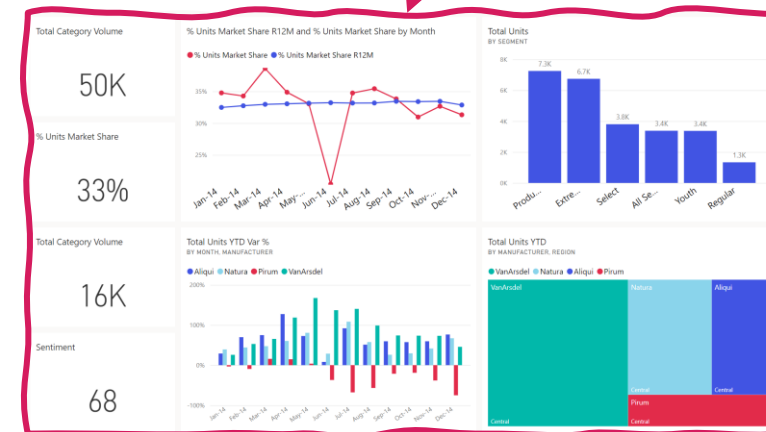
Getting Started Data Analytics



High Level

Data Analytics

To create a report or dashboard, we need to follow the following tasks.



Tasks of a Data Analyst

To create a report or dashboard, we need to follow the following tasks.



Tasks

01

Prepare

cleaning, and transforming your data to get it ready to model and visualize

02

Model

relationships between the tables

03

Visualize

a graphical representation of data

04

Analyze

uncover insights.
DAX

05

Manage

management Power BI assets

THE POWER BI ROADMAP

5 Things Every Every Data Analyst Does

From raw data to real insight.

Prepare

Clean and transform your data to get it ready to model and model and visualize.

01

Model

Build relationships between the tables.

02

Visualize

Create a graphical representation of the data.

03

Analyze

Uncover insights with DAX.

04

Manage

Manage your Power BI assets.

05

5 Steps. One Workflow.

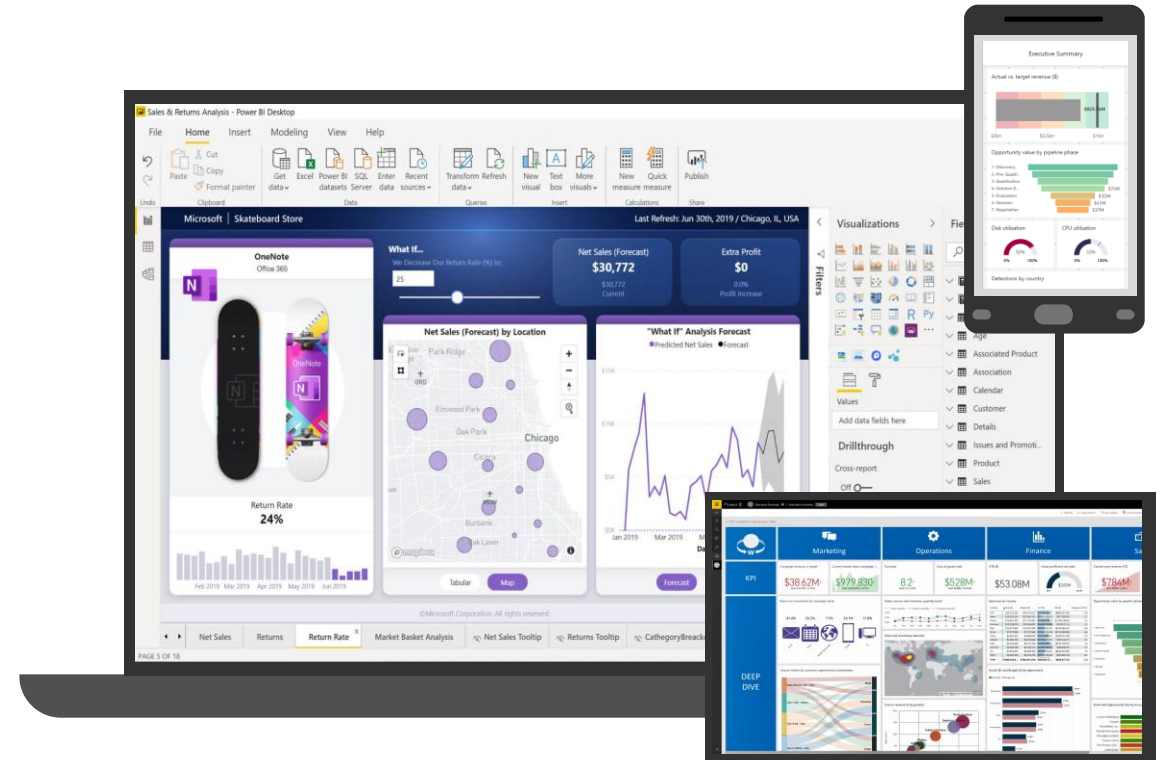
Let's get started.



Introduction to Power BI

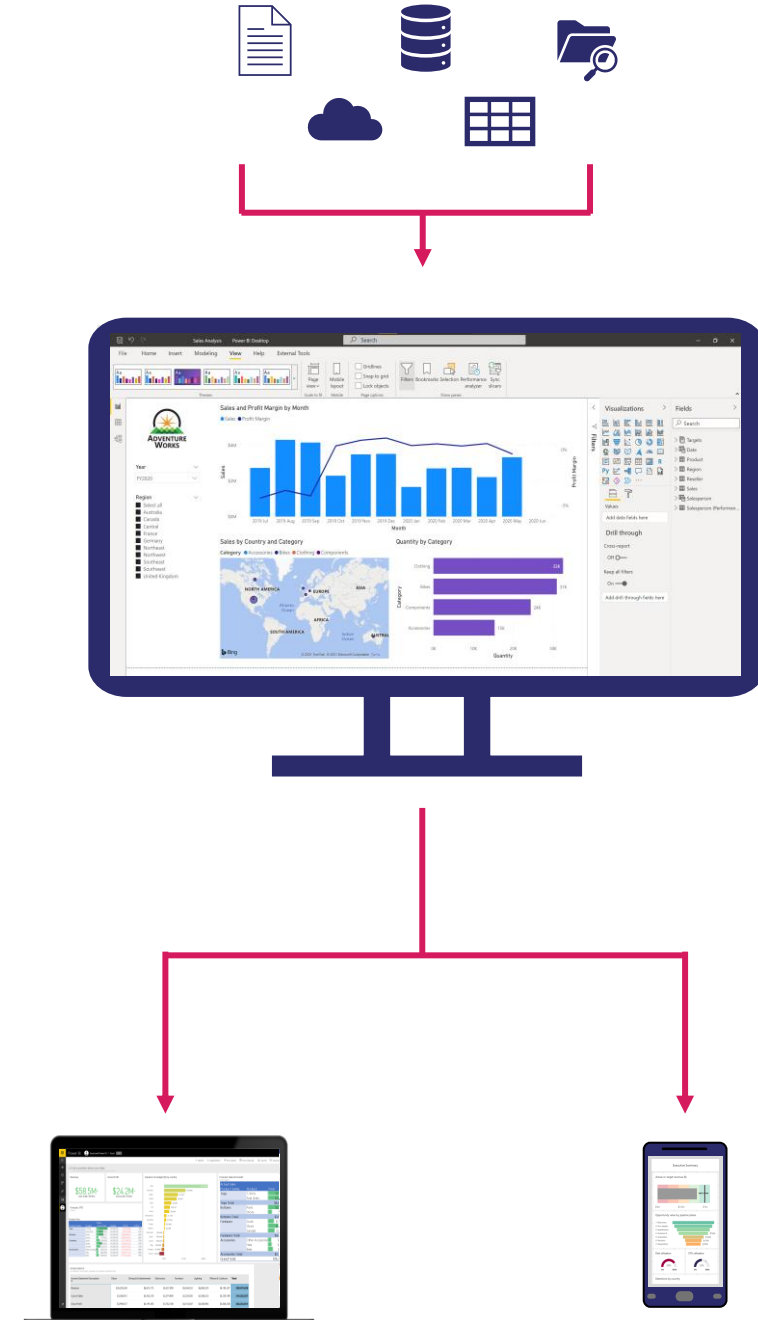
Power BI

A collection of software, services, apps, and connectors.



Common Flow of work in Microsoft Power BI

1. Get data into the Power BI where the data is cleaned, transformed, and modeled, and reports are created.
2. The reports are published to the Power BI service where you build dashboards or create new visualizations.
3. Share dashboards and other Power BI assets with others.
4. View and interact with shared reports and dashboards in the Power BI mobile apps or browsers.



Chapter 01

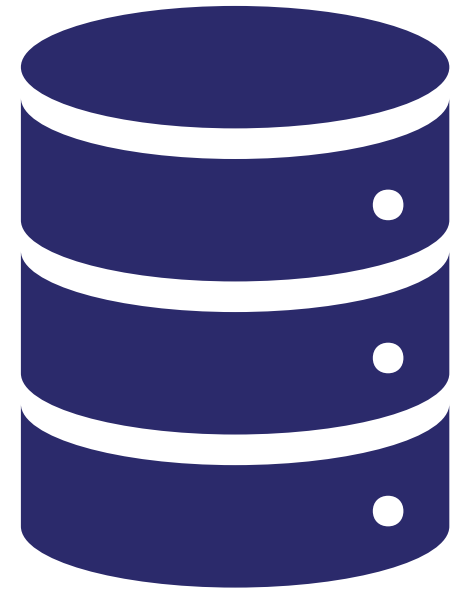
Connect & Transform

Import from Excel, SQL & web. Clean data with Power Query.

CHAPTER 01



Get Data in Power BI



Preparing Training Environment

Lab 0:

Install Power BI Desktop

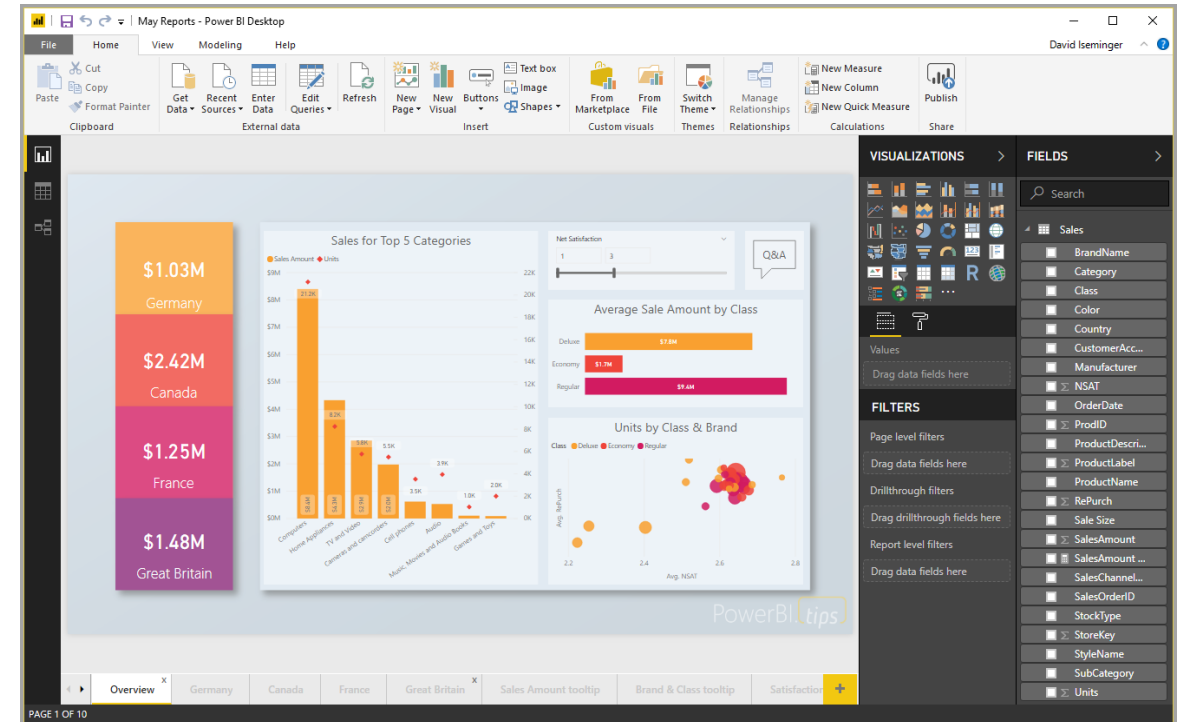
<https://powerbi.microsoft.com/en-us/downloads/>

Create a free Office 365 Account

<https://www.microsoft.com/en-us/microsoft-365/enterprise/office-365-e5>

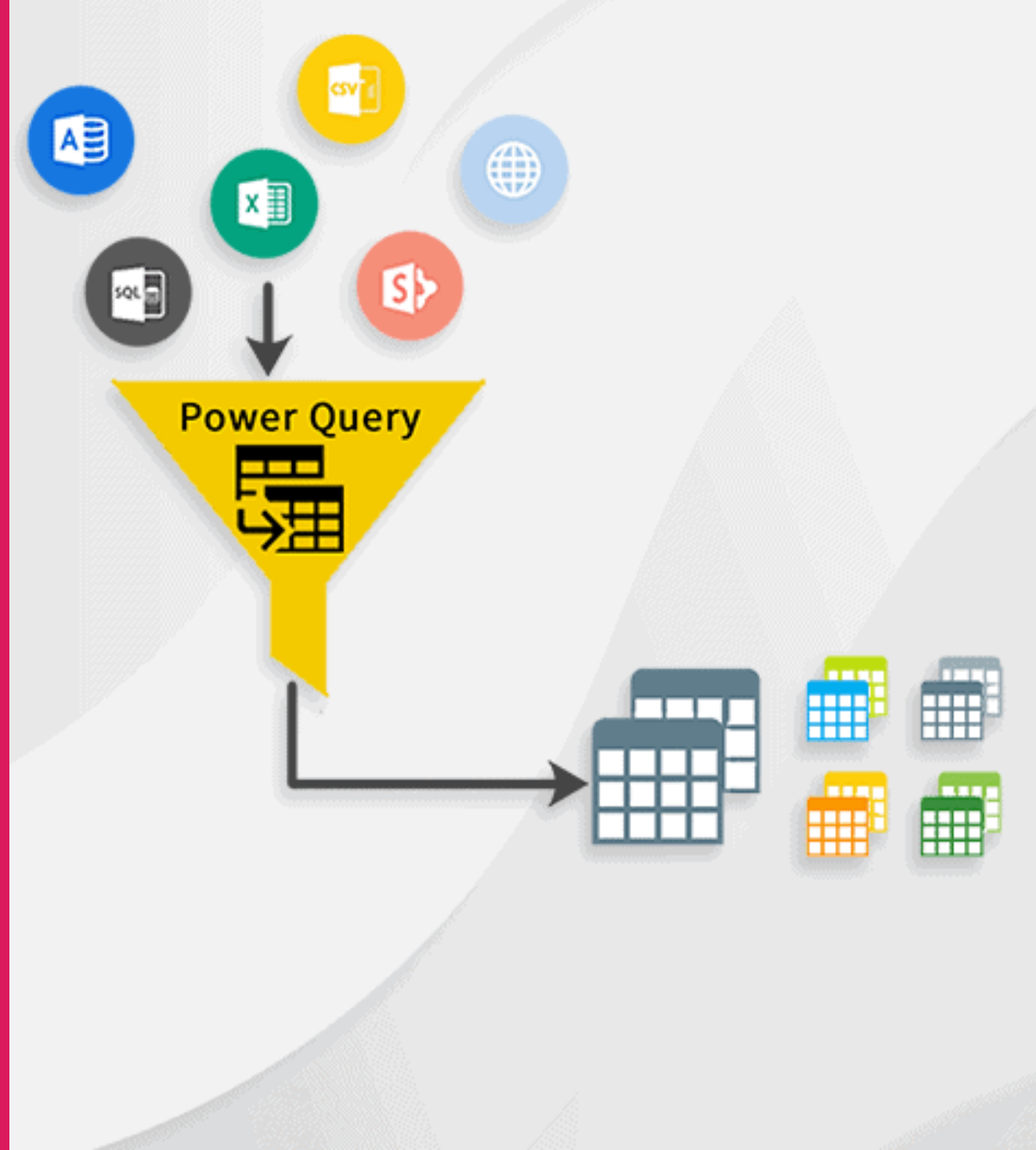
Download the lab files

<https://github.com/sbeeh-software/BI>



Introduction to Getting Data

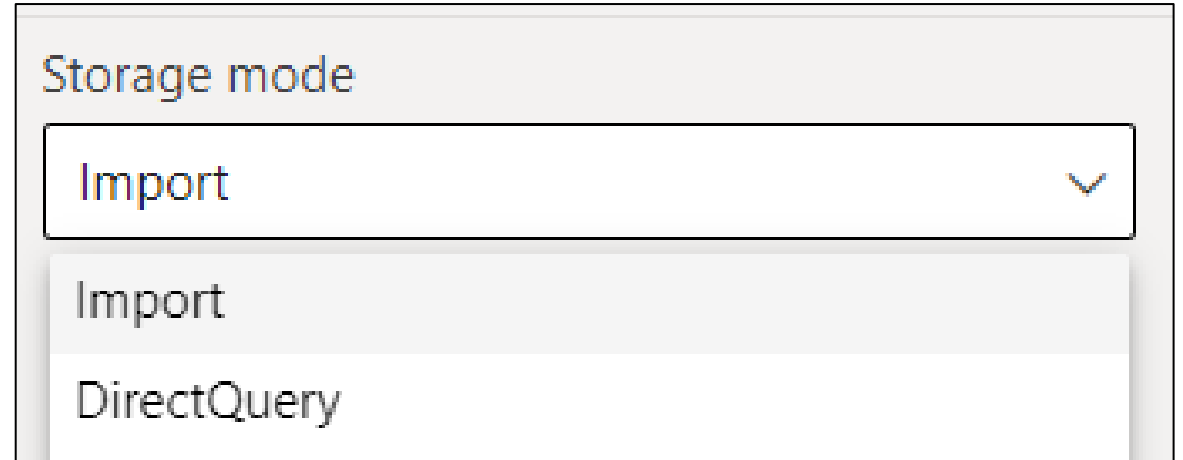
The first step in the data analysis process is identifying and getting data.



Storage Mode

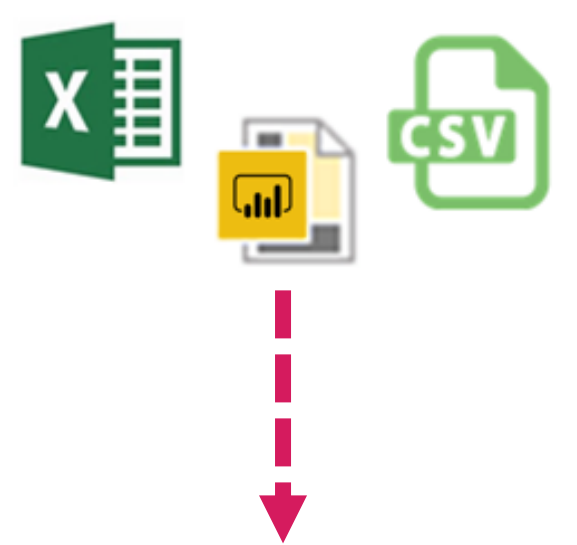
Storage mode affects:

- Available transformations
- Report performance



Lab 01

Get data from flat files



ResellerSales_202006.csv

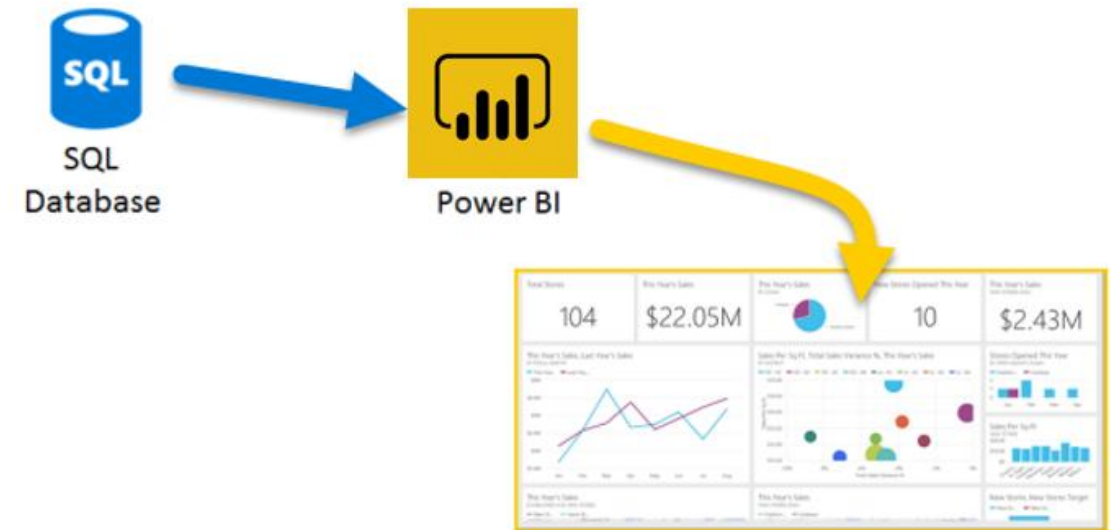
File Origin: 1200; Unicode | Delimiter: Comma | Data Type Detection: Based on first 200 rows

SalesOrderNumber	SalesOrderLineNumber	OrderDate	DueDate	ShipDate	ProductKey	ResellerKey	PromotionKey
SO71691	2	6/1/2020	6/11/2020	6/8/2020	434	104	
SO71691	4	6/1/2020	6/11/2020	6/8/2020	222	104	
SO71774	1	6/1/2020	6/11/2020	6/8/2020	436	609	
SO71774	2	6/1/2020	6/11/2020	6/8/2020	418	609	
SO71775	1	6/1/2020	6/11/2020	6/8/2020	573	595	
SO71775	2	6/1/2020	6/11/2020	6/8/2020	555	595	
SO71775	3	6/1/2020	6/11/2020	6/8/2020	490	595	
SO71776	1	6/2/2020	6/12/2020	6/9/2020	514	106	
SO71777	1	6/2/2020	6/12/2020	6/9/2020	408	128	
SO71777	2	6/2/2020	6/12/2020	6/9/2020	436	128	
SO71778	1	6/2/2020	6/12/2020	6/9/2020	467	557	
SO71778	2	6/2/2020	6/12/2020	6/9/2020	566	557	

Load Transform Data Cancel

Lab 02

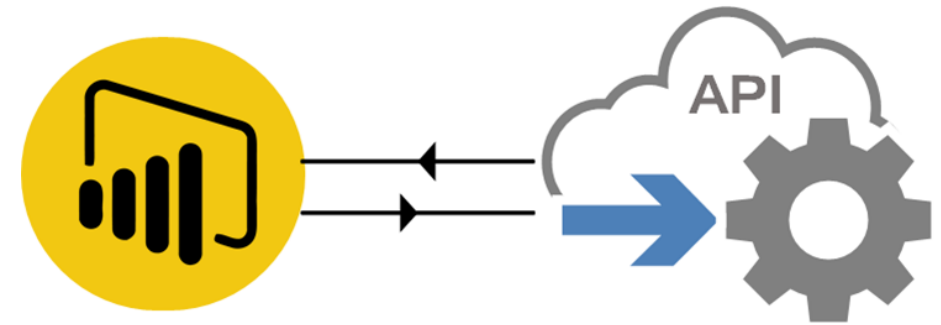
Get data from relational data sources



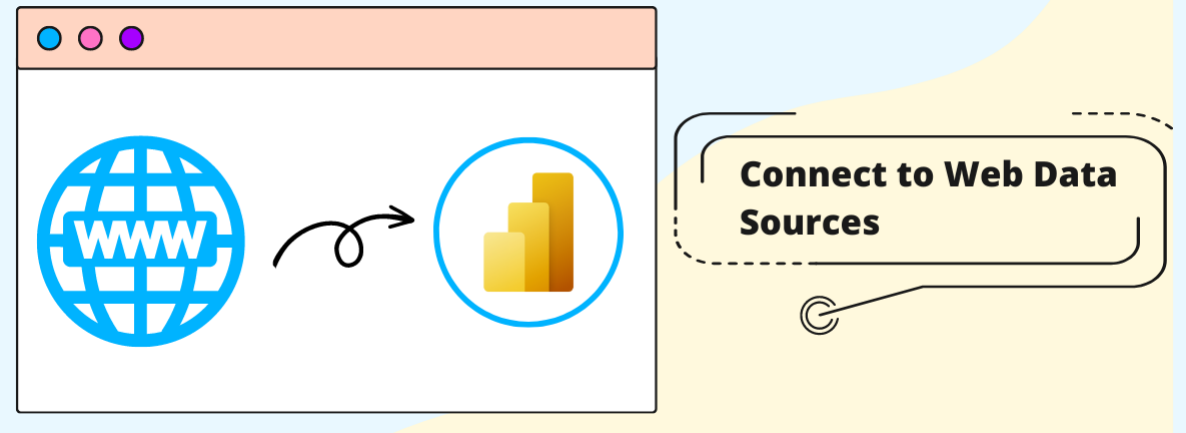
Lab 03

Get data from Service / API

Power BI supports connecting to APIs.



Connect Website



Data Profiling

Data profiling is the process of understanding more about the data.



Lab 05

Data Profiling in Power BI

Monospaced

Show Whitespace

Column quality

Column distribution

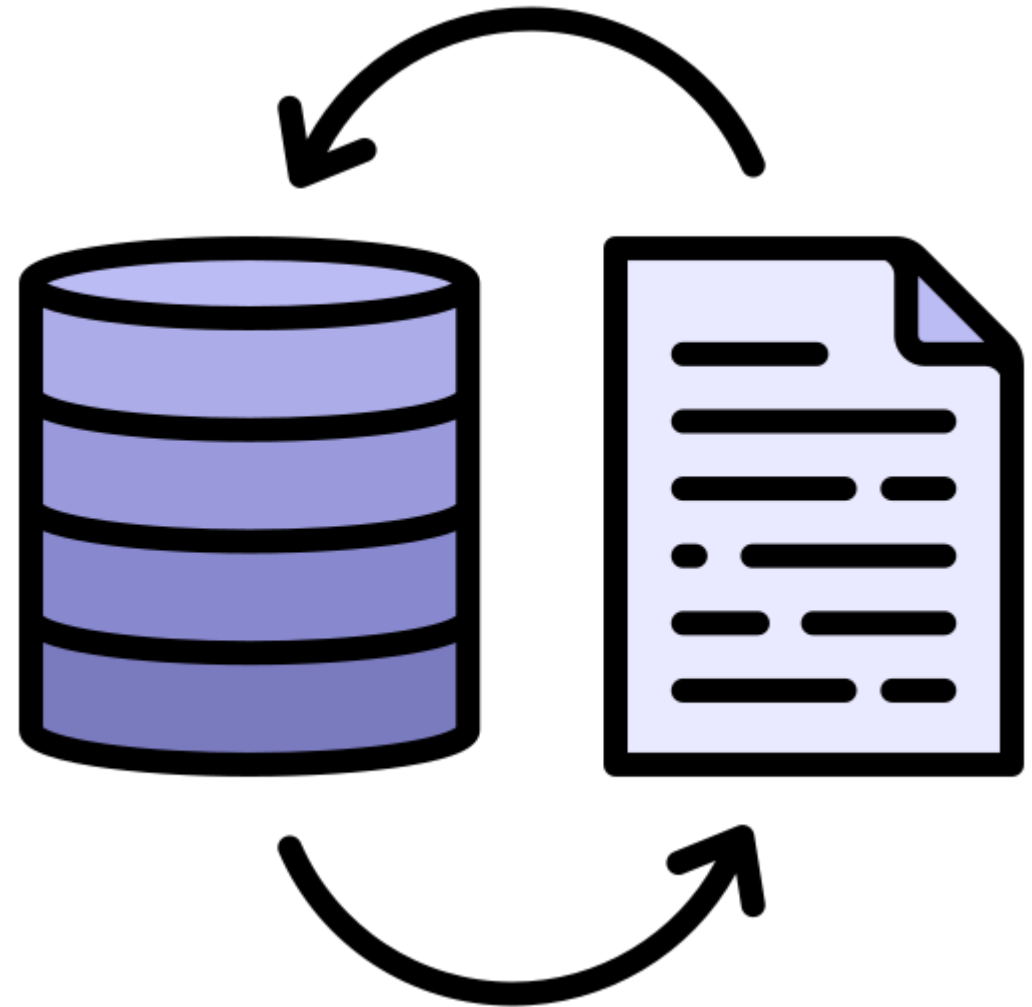
Column profile



Data Transformation

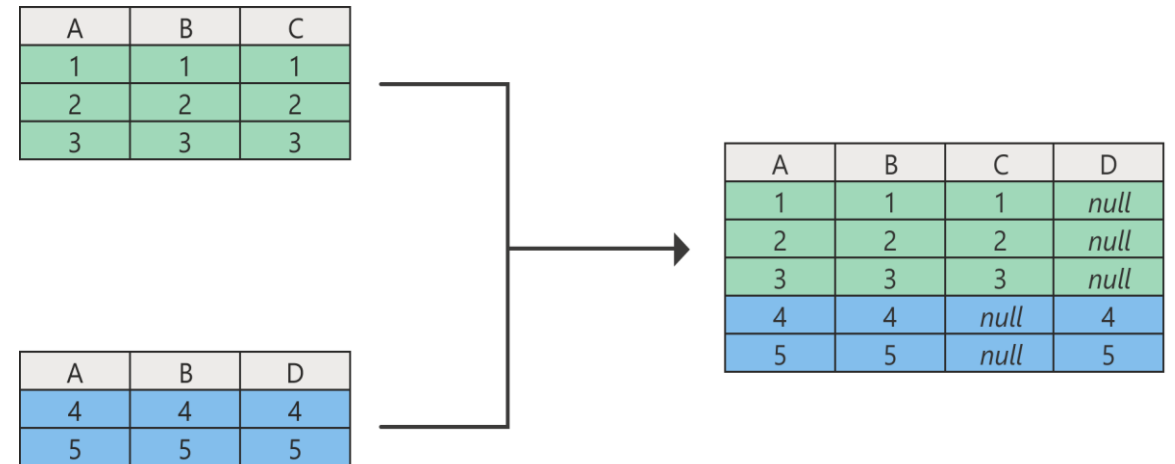
Data Transformation is the process of converting data from one format, structure, or value set into another format so it can be used for analysis, reporting, integration, or storage.

Clean and transform the data



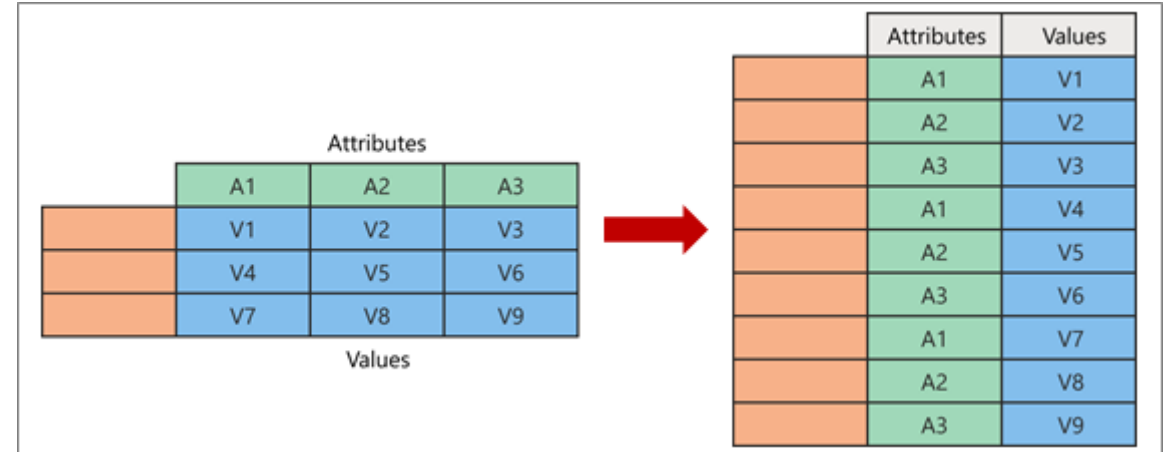
Power Query - Append

Power Query, Append is an operation used to combine two or more tables by stacking them on top of each other.



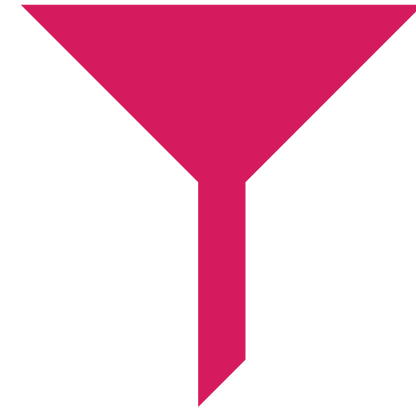
Power Query - Unpivot

Unpivot is a Power Query operation that converts columns into rows.

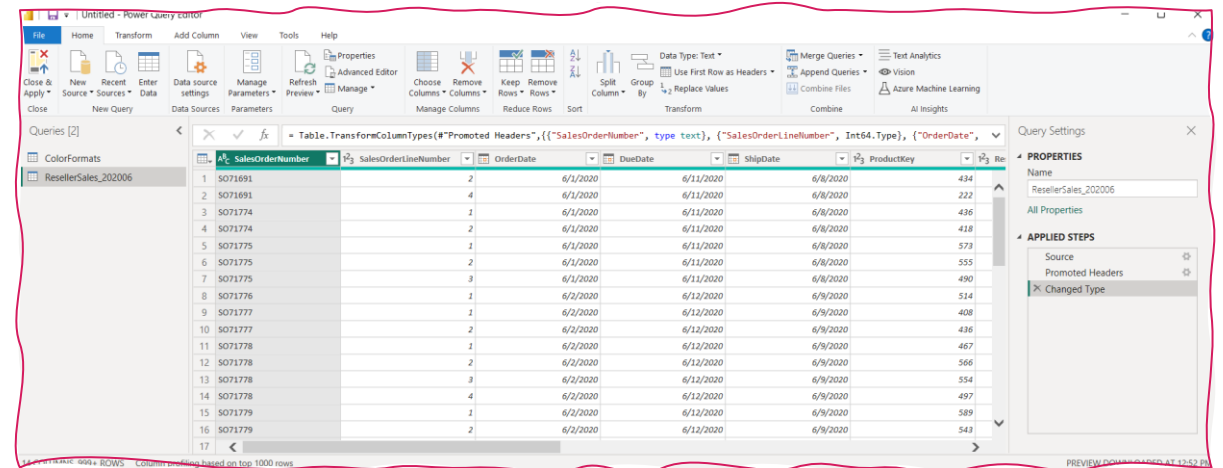


Clean and transform the data

Examples

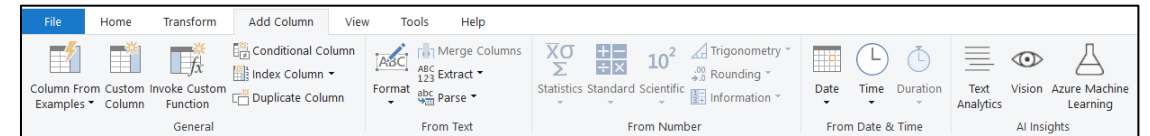
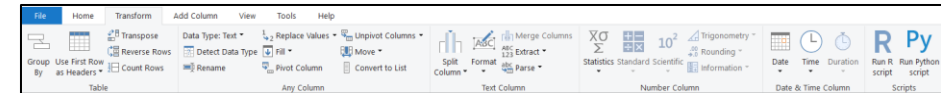


Transform data with Power Query Editor

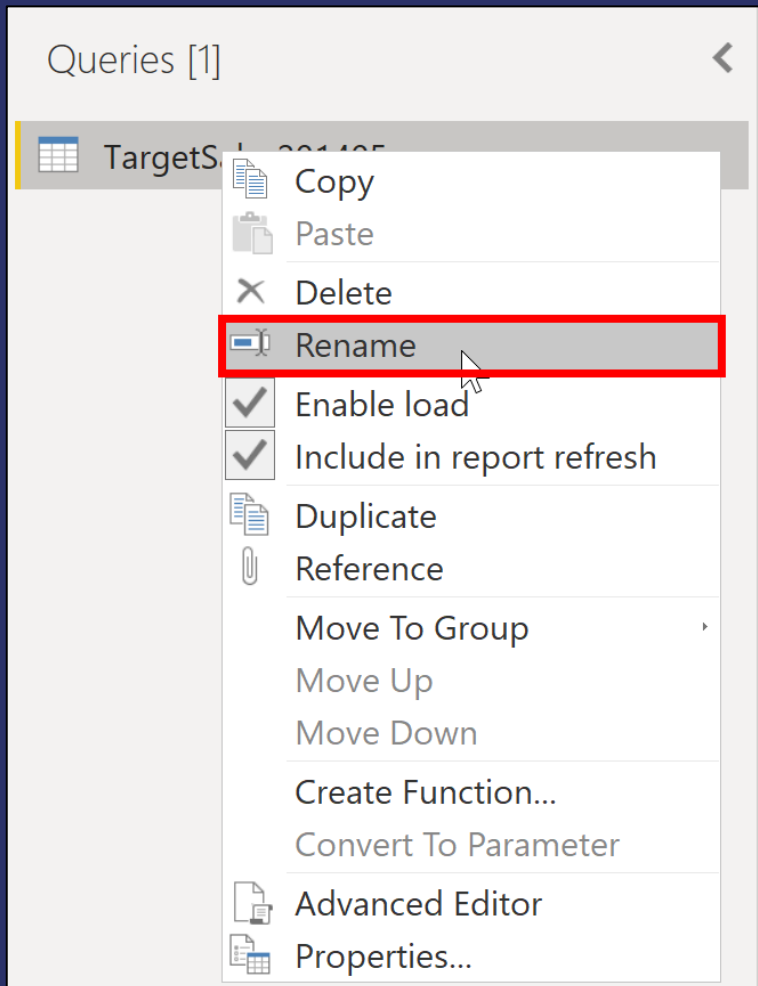


Common transformations

Transform columns, add new, split, extract, and more



User-friendly values



A screenshot of the 'Transform' tab in a data tool. The 'Replace Values' option in the ribbon is highlighted with a red rectangle. Below the ribbon, a table is displayed with columns 'ID', 'Subcategory Name', and 'Attribute'. The 'Attribute' column is highlighted with a yellow background and a red border.

	ID	Subcategory Name	Attribute
1	1	Mountain Bikes	January
2	1	Mountain Bikes	February
3	1	Mountain Bikes	March
4	1	Mountain Bikes	April
5	1	Mountain Bikes	May
6	1	Mountain Bikes	June
7	1	Mountain Bikes	July
8	1	Mountain Bikes	August
9	1	Mountain Bikes	September
10	1	Mountain Bikes	October
11	1	Mountain Bikes	November
12	1	Mountain Bikes	Dezember
13	2	Road Bikes	January
14	2	Road Bikes	February

Shaping table structure

Advanced Editor

Manage ▾

Query

Remove Columns

Remove Other Columns

Choose Columns ▾

Remove Columns ▾

Keep Rows ▾

Remove Rows ▾

Reduce Rows

Sort

	A ^B C Column13	A ^B C Column14
	November	December
	880000	890000
	9500	10000
	511000	512000

Properties

Advanced Editor

Manage ▾

Query

Choose Columns ▾

Remove Columns ▾

Keep Rows ▾

Remove Rows ▾

Remove Top Rows

Remove Bottom Rows

Remove Alternate Rows

Remove Duplicates

Remove Blank Rows

Remove Errors

rs",{{"Month", "Su
January
80000
790000

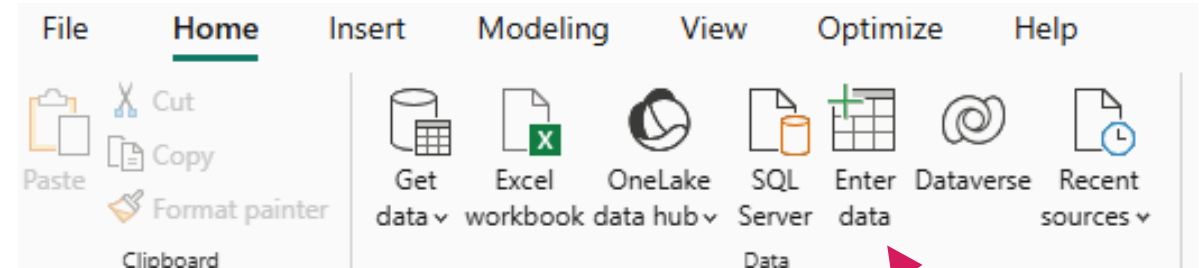
Evaluate and change column data types

ABC SalesOrderNumber	ABC OrderDate	ABC 123 TotalCost
SO43659	7/1/2017	2024.99
SO43659	7/1/2017	6074.97
SO43659	7/1/2017	2024.99
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	4079.98
SO43659	7/1/2017	2039.99
SO43659	7/1/2017	86.52
SO43659	7/1/2017	28.84
SO43659	7/1/2017	34.2

Enter Data

The Enter Data option in Power BI allows you to manually create a new table by directly entering data into Power BI.

Enter Data Option



Main Lab

Sales Report

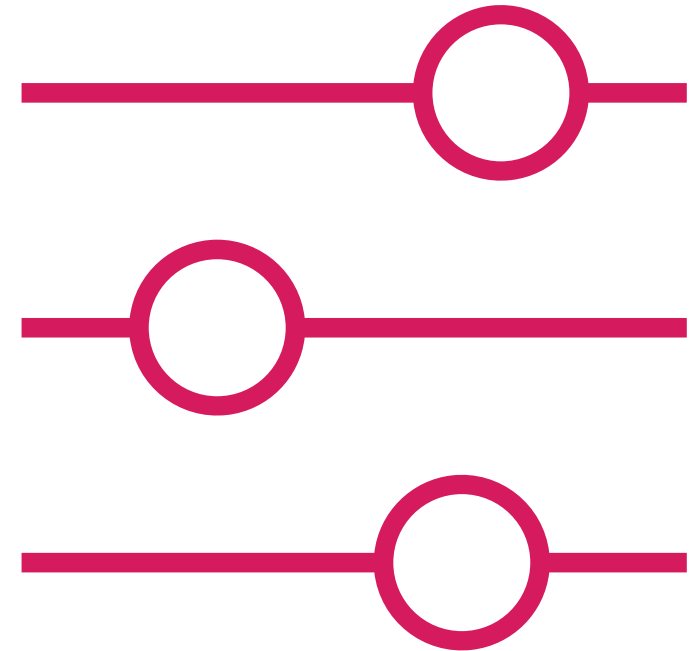
We assume that a company named Contoso has a sales team. The sales team uses an Excel sheet to record all the projects they are working on.

Contoso Case Study



Main Lab Requirements

- What is the total income for all years?
- How is the total income distributed across different years?
- How is the total income distributed across years and months?
- Which months do not have any income recorded?
- Which year has the highest total income?
- Which month has the highest total income in the selected year?
- Which year has the lowest total income?
- Which month has the lowest total income in the selected year?
- How is the total income distributed across different clients?
- Which client is the top performer based on total income?
- Which country is the top performer based on total income?
- What is the number of projects in the selected year?
- What is the number of open projects for the current year?



Additional Challenges

1. Management has asked us to display the full country name instead of the ISO code.
2. Management has requested that we use the full month's name instead of the month's number.
3. We need to display them all months, regardless of whether they contain data or not.
4. For months without data, we need to display the total amount as zero.

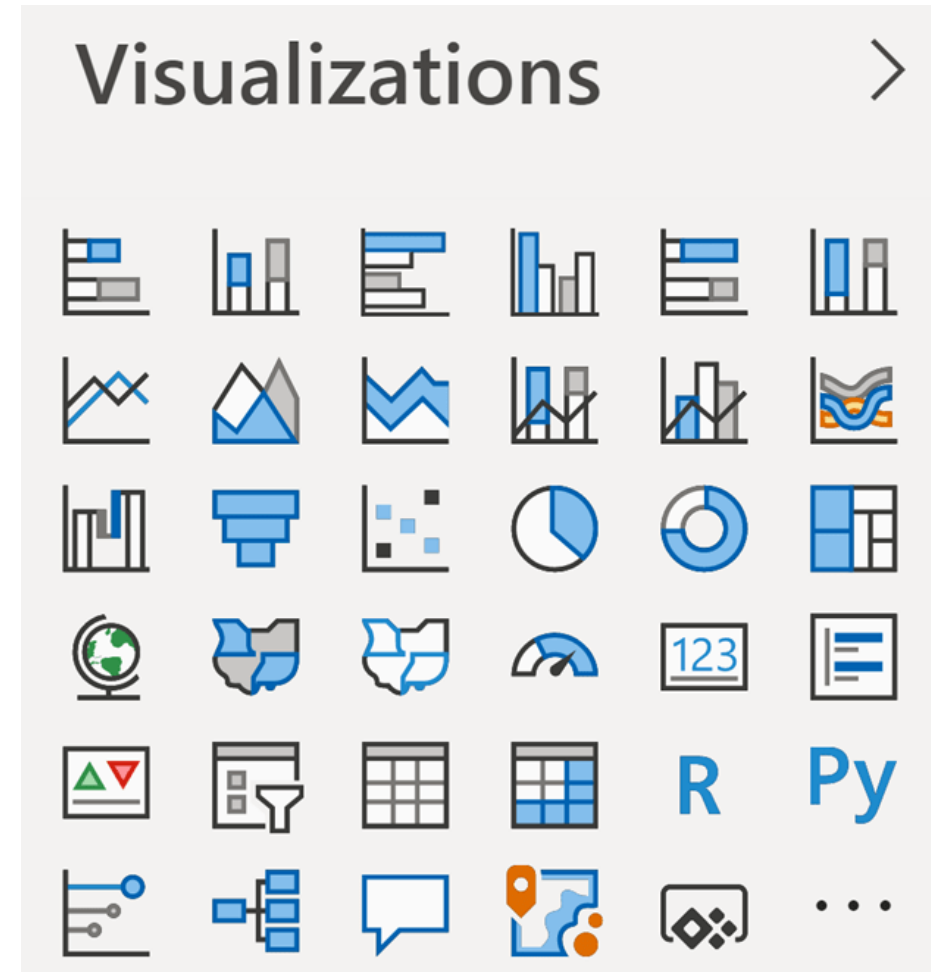


Chapter 02

Visualization

Visualization in Power BI is the process of presenting data graphically to help users understand patterns, trends, relationships, and insights quickly.

CHAPTER 02



Create Report

Main Lab



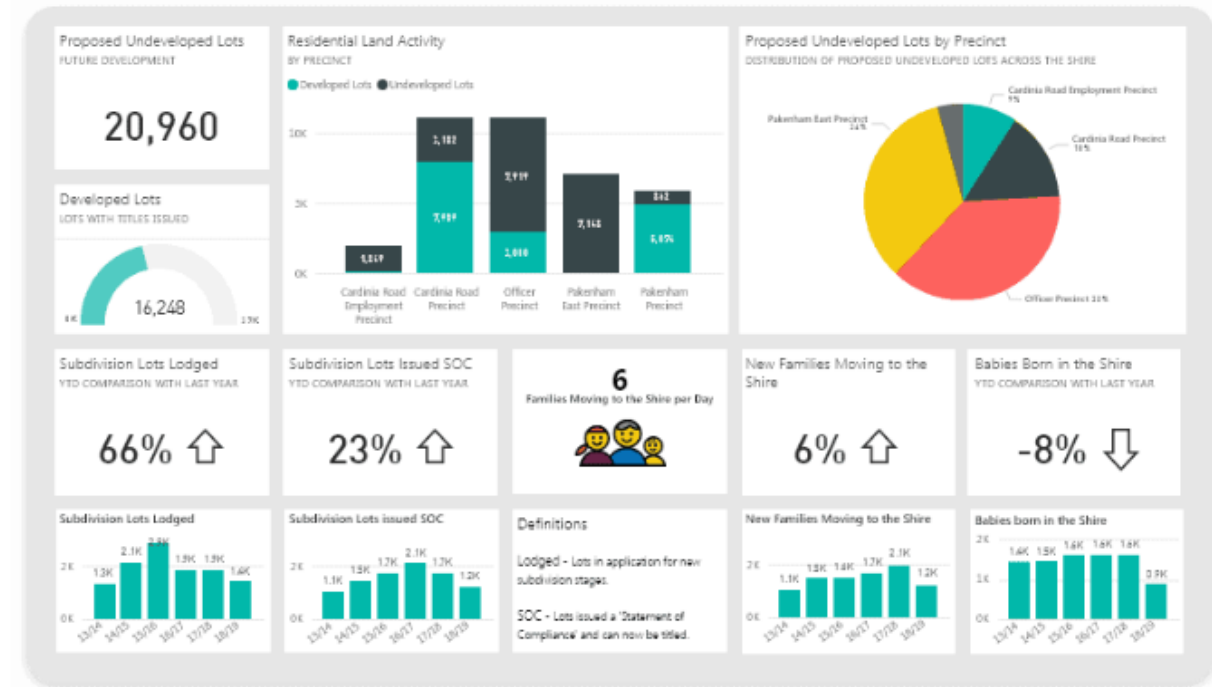
Design a Report

Main Lab



Design a Report Layout

Main Lab



Design a Report Layout

Revenue by Country

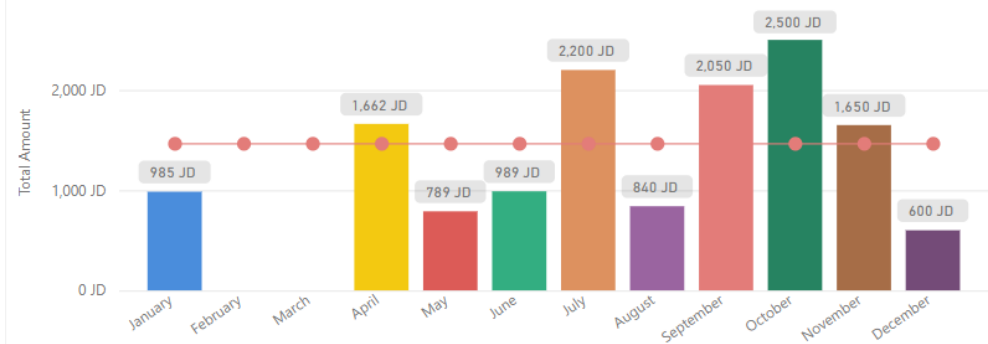
Jordan 13,425 JD



Oman 840 JD



Revenue by Month



Revenue Per Month

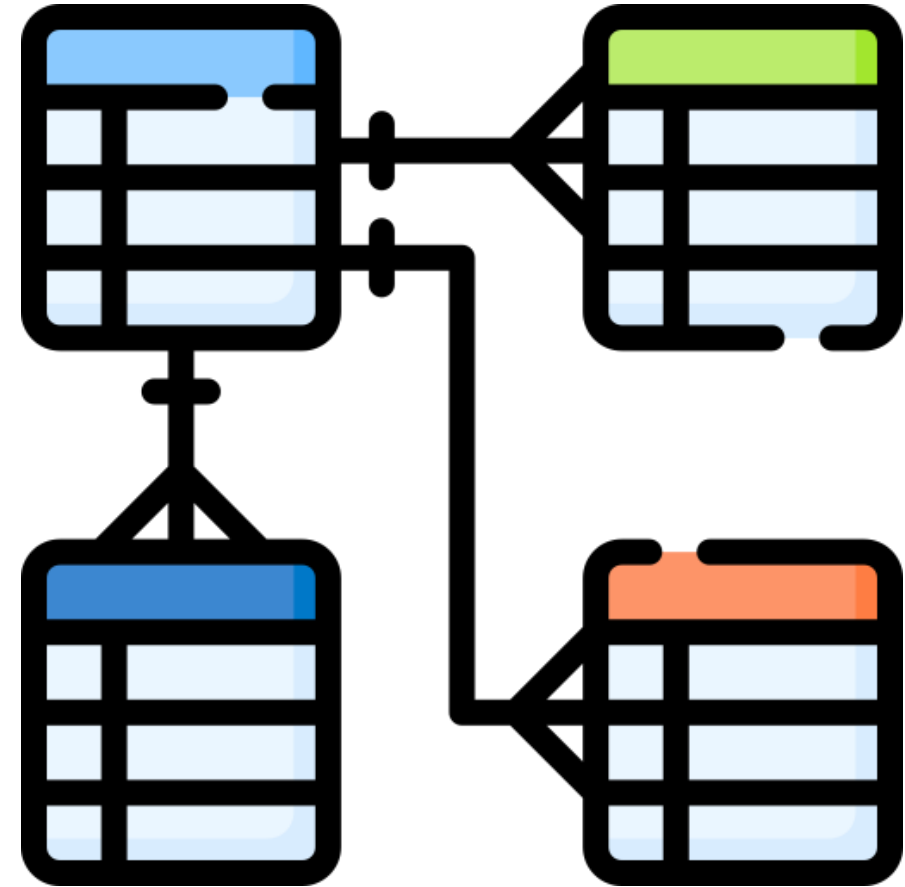
Month	Total Amount	% Total
January	985 JD	5.61%
February	0 JD	0.00%
March	0 JD	0.00%
April	1,662 JD	9.47%
May	789 JD	4.49%
June	989 JD	5.63%
July	2,200 JD	12.53%
August	840 JD	4.78%
September	2,050 JD	11.68%
October	2,500 JD	14.24%
November	1,650 JD	9.40%
December	600 JD	3.42%
Total	14,265 JD	81.25%

Chapter 03

Data Modeling

Power BI Data Modeling refers to the process of creating logical relationships between different tables of data to establish a structure that supports meaningful analysis and reporting.

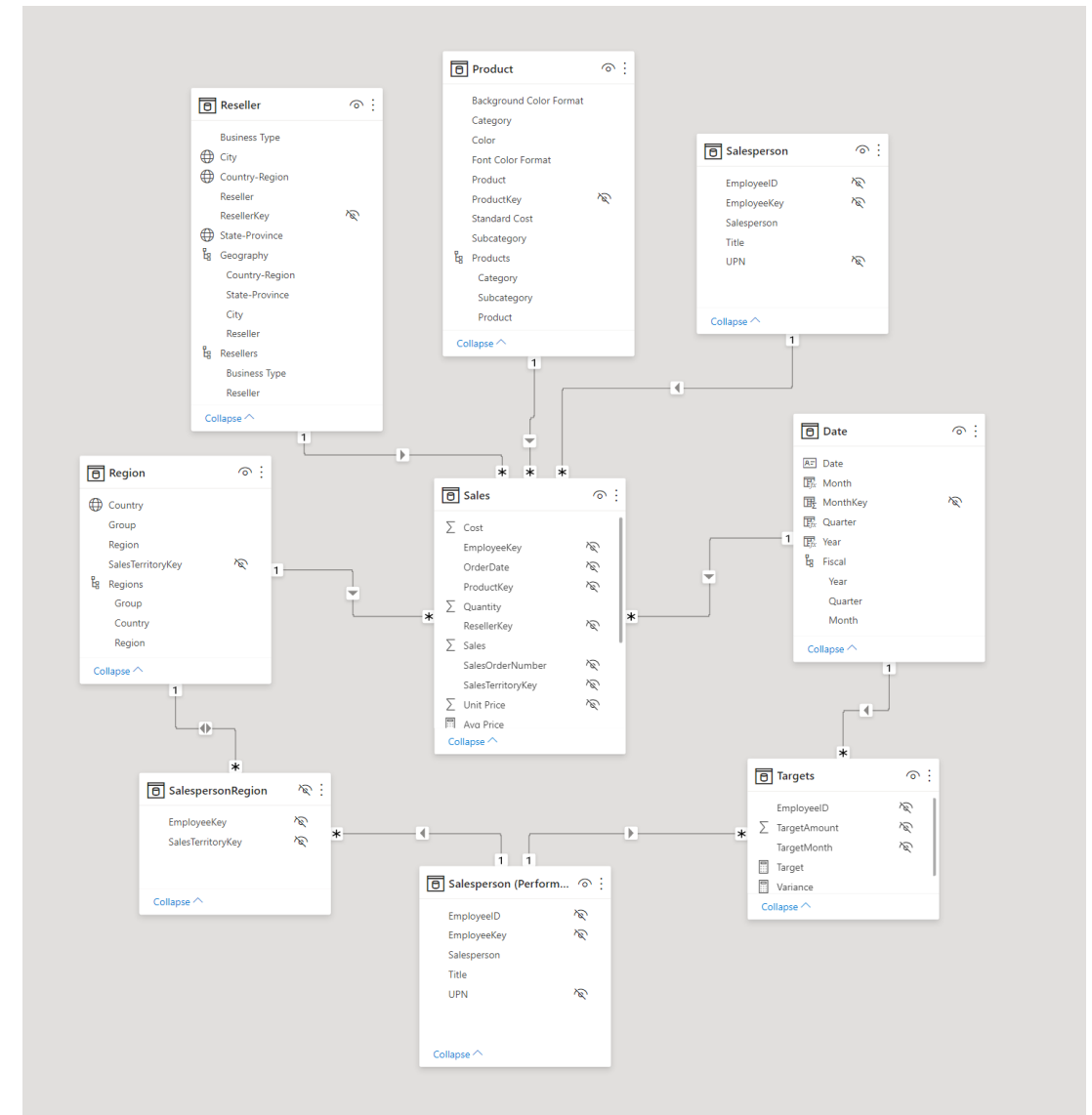
CHAPTER 03



Introduction to Data Modeling

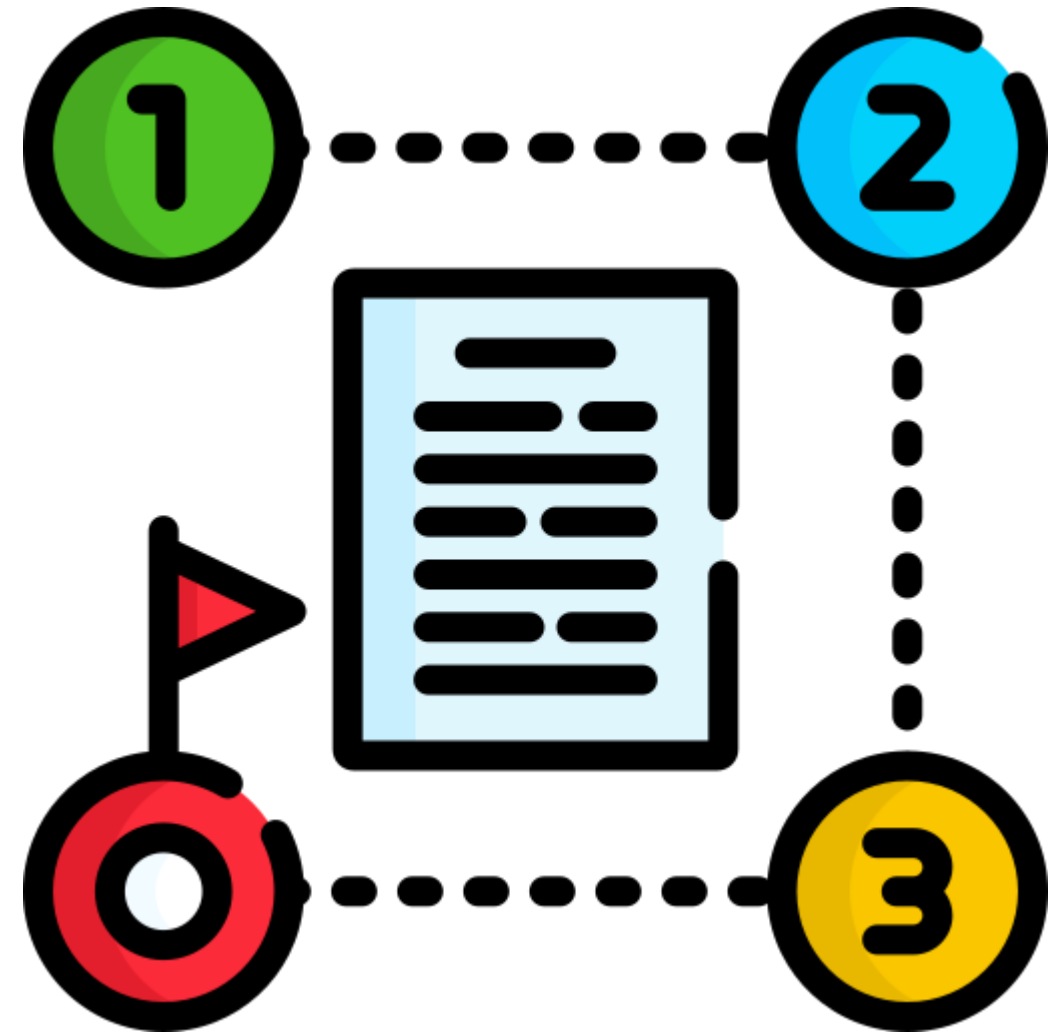
Benefits of a good data model:

- Accurate reports.
- Faster data exploration.
- Simpler aggregations.
- Easier to maintain.



Steps to Create a Data Model in Power BI

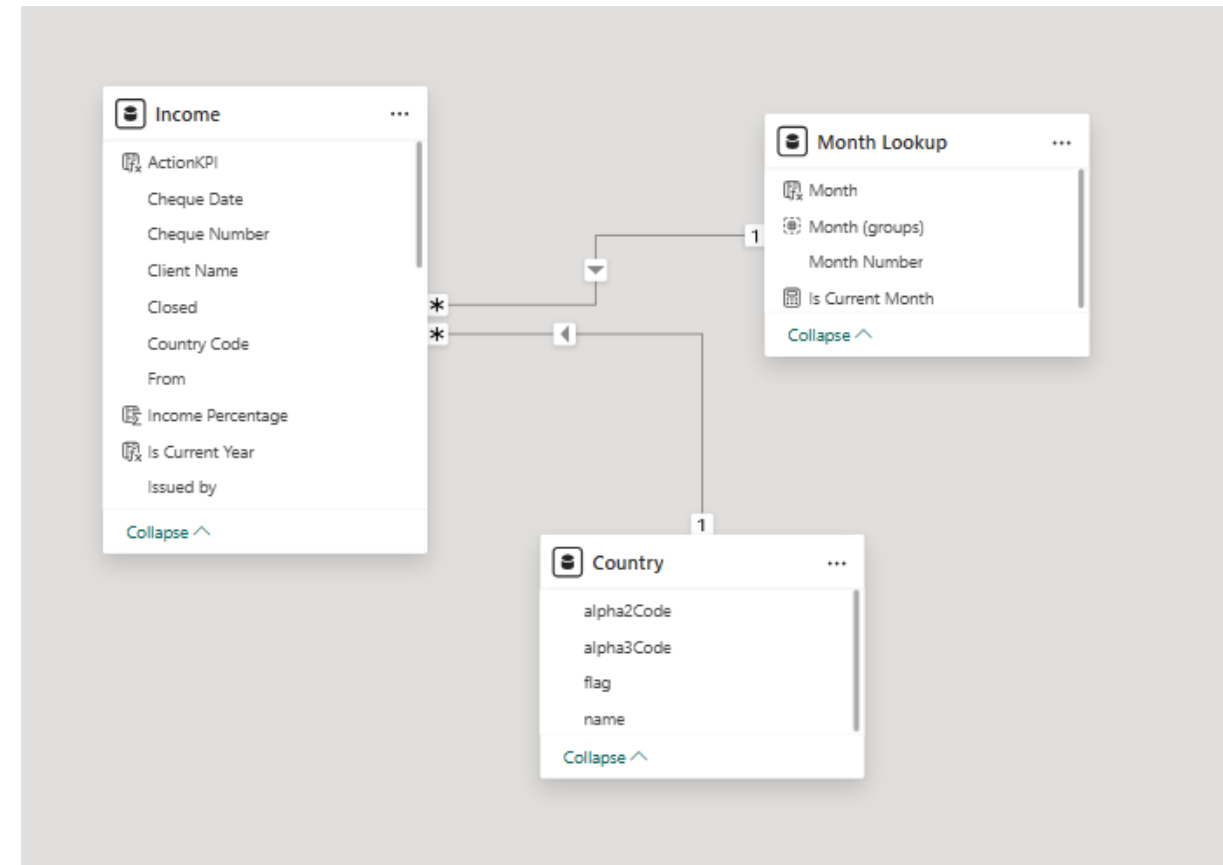
1. Import Data:
2. Load data from different sources such as Excel, databases, or APIs.
3. Clean and Transform Data:
4. Use Power Query to clean and prepare the data (e.g., remove duplicates, filter rows, etc.).
5. Define Relationships:
6. Use the Model View in Power BI to define relationships between tables.
7. Drag and drop fields to connect tables based on common keys.
8. Configure Relationships:
9. Set relationship properties:
10. Validate the Model:
11. Test relationships and calculations to ensure the model works as expected.



Main Lab:

Data Modeling

Implement Data modeling in sales report.



Chapter 03

Data Analysis Expressions

DAX (Data Analysis Expressions) is a formula language used in Power BI to create custom calculations and data analysis expressions. It is similar to Excel formulas but more powerful for working with relational data models.

CHAPTER 04



Introduction to DAX

What is DAX in Power BI?

DAX (Data Analysis Expressions) is a formula language used in Power BI to create custom calculations and data analysis expressions. It is similar to Excel formulas but more powerful for working with relational data models.

Example:

Calculated Columns – Create new columns based on existing data.

Measures – Define aggregations like SUM, AVERAGE, or COUNT.



Examples

Basic Aggregation

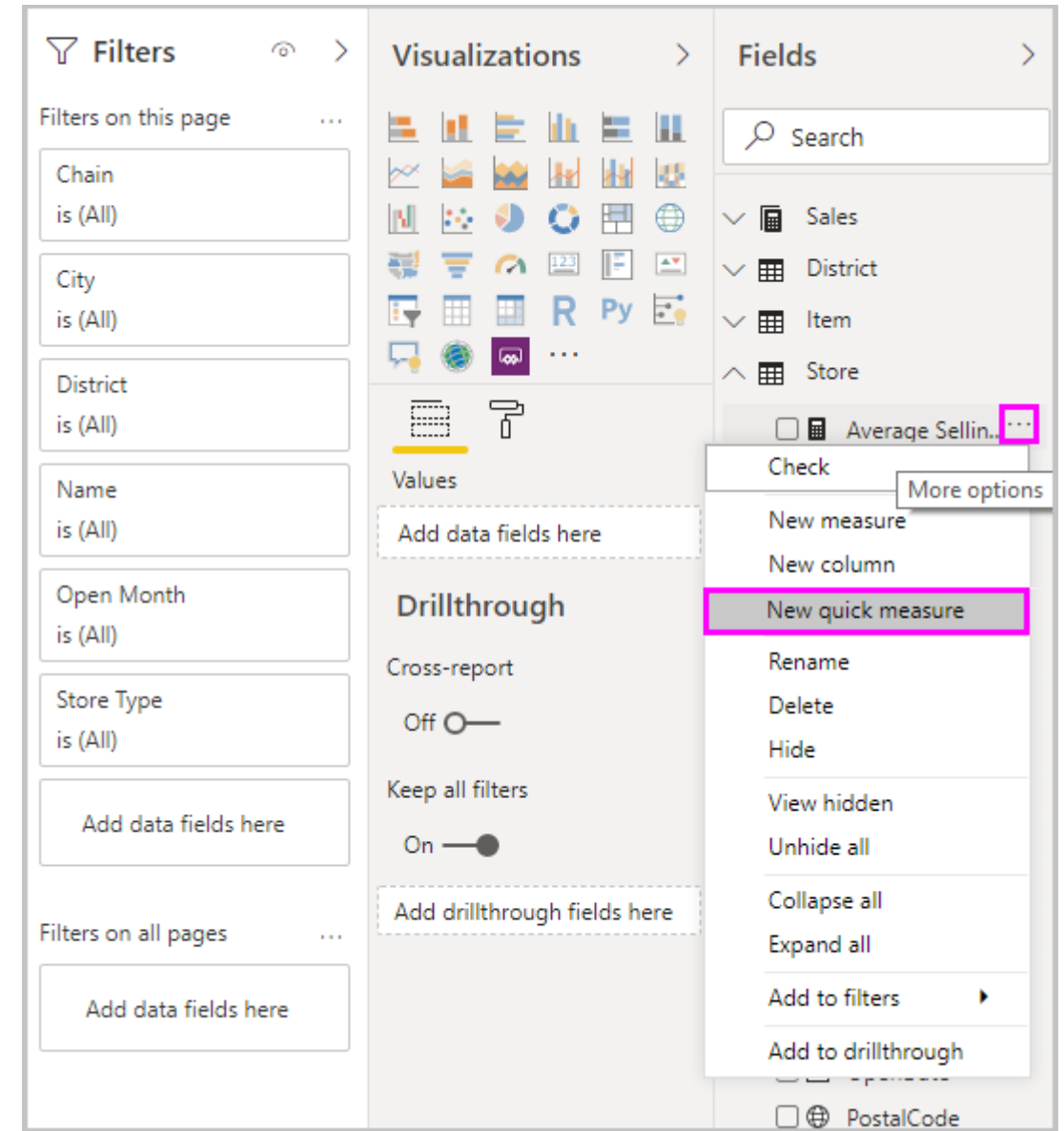
Total Sales = SUM(Sales[Revenue])

Conditional Calculation

High Value Sales = IF(SUM(Sales[Revenue]) > 5000,
"High", "Low")

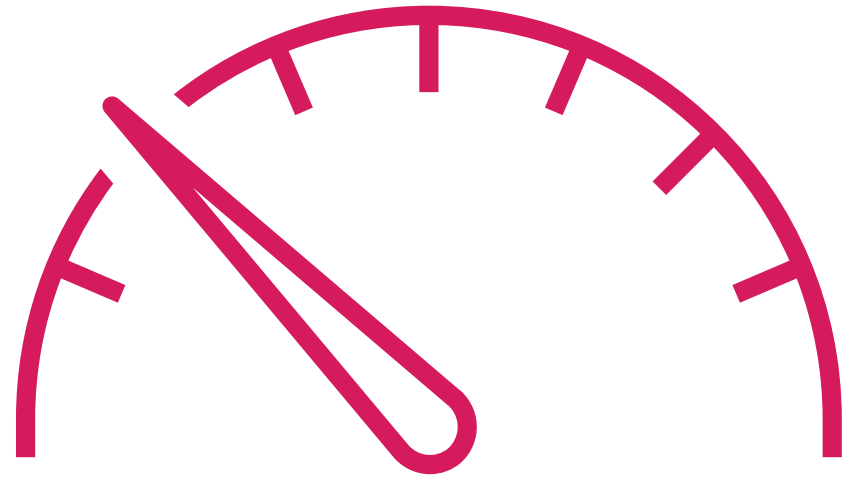
Measures

Measures are a summarization of data.
A way of defining aggregate calculations on data.
Often called "Calculated Measures".



Create Measures

Main Lab



Calculated Columns

Total Price = 'Sales OrderDetails'[Quantity] * 'Sales OrderDetails'[Unit Price]

Structure		Formatting		Properties	Sort
1		Column =			
10248	11	12	\$14		
10248	42	10	\$9.8		
10248	72	5	\$34.8		
10249	14	9	\$18.6		
10249	51	40	\$42.4		
10250	41	10	\$7.7		

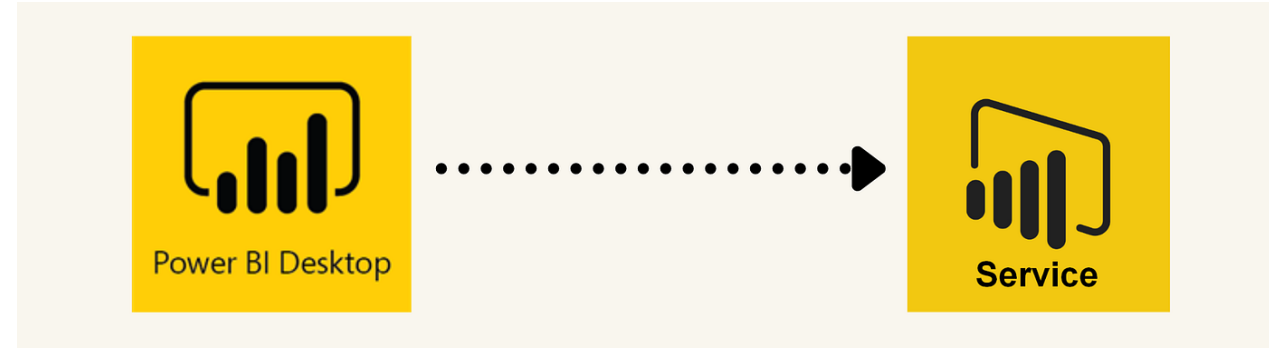
Order ID	Product ID	Quantity	Unit Price	Total Price
10248	11	12	\$14	\$168
10248	42	10	\$9.8	\$98
10248	72	5	\$34.8	\$174
10249	14	9	\$18.6	\$167.4
10249	51	40	\$42.4	\$1,696
10250	41	10	\$7.7	\$77
10250	51	35	\$42.4	\$1,484
10250	65	15	\$16.8	\$252
10251	22	6	\$16.8	\$100.8
10251	57	15	\$15.6	\$234
10251	65	20	\$16.8	\$336
10252	20	40	\$64.8	\$2,592
10252	33	25	\$2	\$50
10252	60	40	\$27.2	\$1,088

Chapter 05

Publishing & Management Power BI assets

Publishing & Management in Power BI refers to deploying, sharing, securing, refreshing, and maintaining Power BI content after development.

CHAPTER 05



Publishing a Report

1. Create report in Power BI Desktop.
2. Save the .pbix file.
3. Select Publish.
4. Sign in to Power BI.
5. Choose a workspace.
6. Click Publish.



Power BI Row-Level Security (RLS)

Row-Level Security (RLS) in Power BI is a feature that allows you to restrict access to data at the row level, meaning different users can see different subsets of data based on their permissions or roles.

Static Method

Uses a fixed value in the DAX filter.

Manage roles

Roles

Automotive ...

Clothing ...

Game ...

Sports ...

Create Delete

Tables

Employees ...

Sales ...

Table filter DAX expression

[department] = "Game"

Filter the data that this role can see by entering a DAX filter expression that returns a True/False value. For example: [Entity ID] = "Value"

Save Cancel

Dynamic Method

Uses a dynamic value in the DAX filter.

Manage roles

Roles

EmployeeEmailAddress ...

Create

Delete

Tables

Employees ...

Products ...

Sales ...

Table filter DAX expression

[emailAddress] = userprincipalname()

<https://www.metricsthatmatter.com/url/u.aspx?459AD5B65239962787>

Thank you for your attention.
Questions and Answers.

SBEEH
Software

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www.sbeeh.com



Mohammad Sbeeh

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